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Mechanical impact behavior of polyether–ether–ketone (PEEK)

D. Garcia-Gonzalez^a, A. Rusinek^b, T. Jankowiak^c, A. Arias^{a,*}

^a Department of Continuum Mechanics and Structural Analysis, University Carlos III of Madrid, Avda. de la Universidad 30, 28911 Leganés, Madrid, Spain

^b Laboratory of Mechanics, Biomechanics, Polymers and Structures, National Engineering School of Metz, 1 route d'Arts Laquenexy, CS 65820 57078 METZ Cedex 3, France

^c Institute of Structural Engineering, Poznan University of Technology, Piotrowo 5, Poznan, Poland

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This paper deals with the mechanical behavior of polyether ether ketone (PEEK) under impact loading. PEEK polymers are the great interested in the field of medical implants due to their biocompatibility, weight reduction, radiology advantage and 3D printing properties. Implant applications can involve impact loading during useful life and medical installation, such as hip systems, bone anchors and cranial prostheses. In this work, the mechanical impact behavior of PEEK is compared with Ti6Al4V titanium alloy commonly used for medical applications. In order to calculate the kinetic energy absorption in the impact process, perforation tests have been conducted on plates of both materials using steel spheres of 1.3 g mass as rigid penetrators. The perforation test covered impact kinetic energies from 21 J to 131 J, the equivalent range observed in a fall, an accidental impact or a bike accident. At all impact energies, the ductile process of PEEK plates was noted and no evidence of brittle failure was observed. Numerical modeling that includes rate dependent material is presented and validated with experimental data.

1. Introduction

The developments of new technologies and biocompatible materials have made it possible to replace more parts of the human body. Titanium alloy was generally used [1], but in recent years, following confirmation of biocompatibility [2,3], the polymer poly ether ether ketone (PEEK) has been increasingly employed as matrix material for composites in trauma, orthopedic, dentals, spinal and cranial implants [4–6]. Implant applications of PEEK materials usually involve impact loading during medical installation and the useful life of prostheses, such as hip stems, bone anchors and cranial implants, Fig. 1. In this regard, cranial implants have experienced a significant evolution in the last decade in different aspects such as materials, method of fixation, and manufacturing process [5]. An important aspect to take into account is the load bearing capacity of these structural prostheses. Indeed, the implant must resist at different loading including those generated by a fall, an accidental impact or a bike accident. Mechanical impact process is a complex problem that includes dynamic behavior, fracture, damage, contact and friction [7]. Therefore, impact loads can affect the structural response of materials used in prosthetic devices. In particular, the dynamic behavior of PEEK

composites is highly influenced by the matrix properties due to its semi crystalline nature [8]. About this matter, interesting thermo mechanical phenomena have been reported for PEEK matrix [9,10], including changes in crystallinity, deformation induced heating, macroscopic decolouration, high strain rate and large deformations associated with impact. The elastic properties of PEEK are relatively unaffected by rate effects at body temperature, which is below the glass transition [4]. However, the yielding and plastic flow behaviors are affected by strain rate at physiological temperatures [10,11]. Additionally, adiabatic heating, associated with dynamic behavior of the impact process can induce rapid crystallization of PEEK at temperatures above glass transition [12].

In this regard, the impact behavior of PEEK has not been deeply studied in terms of kinetic energy absorption and failure under impact loading [11,13], and perforation tests have not been reported in the scientific literature.

In this work, perforation tests using rigid spheres have been conducted on plates of PEEK 450G and compared with Ti6Al4V titanium alloy, an alloy frequently used in medical applications. The perforation experiment covered impact kinetic energies from 21 J to 131 J. A numerical approach, including the influence of strain rate and temperature, is presented and validated with experimental data. This numerical model can be useful in the design of prosthetic implants subjected to impact loading.

2. Material

A commercial plate of unfilled PEEK 450G, general purpose grade, with a density $\rho = 1300 \text{ kg m}^{-3}$ was purchased measuring $130 \times 130 \times 30 \text{ mm}^3$. Mechanical and thermal properties are reported in Table 1 [14] in agreement with data published by other authors [11]. In addition to high strength and biocompatibility, unfilled PEEK has been used in both cervical and lumbar spinal cages and cranial implants with considerable clinical success [56] and [15].

2.1. Crystallinity

PEEK is a two phase semi crystalline polymer, consisting of an amorphous phase and a crystalline phase. The crystalline content of injection molded PEEK in implants typically ranges from 30% to 35%, [4]. From differential scanning calorimetry (DSC) a degree crystallinity of $30 \pm 2\%$ was calculated for PEEK 450G by integrating the melt endotherm and relating it to the literature value of 100% crystalline PEEK [16]. These data are in agreement with data reported by El Qoubaa and Ramzi [10] for PEEK 450G and with the calculated degree of crystallinity χ_c , by Eq. (1):

$$\chi_c = \frac{\rho_c(\rho - \rho_a)}{\rho(\rho_c - \rho_a)} \quad (1)$$

where ρ is the sample density, $\rho_a = 1260 \text{ kg m}^{-3}$, is the extrapolated density of the pure amorphous phase and $\rho_c = 1400 \text{ kg m}^{-3}$, is the extrapolated density of pure crystalline phase [16]. Mechanical properties of PEEK materials are influenced by the degree of crystallinity. Several authors have shown that increasing the degree of crystallinity can increase elastic modulus and yield strength while decreasing fracture toughness [11,17,18]. Different behaviors have been reported generating some controversy about the relationship between crystallinity and high strain rate. Hamdan and Swallowe [12] reported an increase in crystallinity of samples deformed by large strains under adiabatic conditions. However, Rae et al. [11] reported a decrease in crystallinity of all samples deformed by large strains.

2.2. Strain rate and temperature sensitivity

The PEEK 450G has been tested by Rae et al. [11] for different initial temperatures and strain rates. In this study a positive strain rate sensitivity was observed with the mechanical properties increasing with the strain rate. However, for a certain strain rate,

Table 1
Material properties of PEEK [14].

Mechanical properties		Thermal properties	
Elastic modulus (GPa)	3.6	Thermal conductivity (W/m K)	0.29
Poisson's ratio	0.38	Specific heat (J/kg K)	2180
Density (kg/m ³)	1300	Glass transition temperature (K)	416
Yield stress (MPa)	107	Melt transition temperature (K)	616

$\dot{\epsilon}_{transition} \approx 0.1 \text{ s}^{-1}$, a thermal softening is observed and therefore, the process of plastic deformation has to be considered as adiabatic, Fig. 2a. Concerning the temperature effect, a loss of ductility was observed when the initial temperature was lower than the room temperature $T_0 = 300 \text{ K}$. Nonetheless, the ductility was retained with a strain level larger than $\epsilon_{low temperature} > 0.2$. In addition, using experimental results published in [11], it was observed that the material behavior was still ductile for a temperature higher than $T_0 > 413 \text{ K}$, Fig. 2b, knowing that the glass temperature is equal to $T_g > 416 \text{ K}$. It was also demonstrated that PEEK behavior was more brittle under tensile loading at low temperature compared with compression loading [11], with a failure strain level lower than $\epsilon_{failure}^{tension} < 0.1$ at 223 K . Overall, the temperature sensitivity and the strain rate sensitivity are similar under tension and compression. The two major parameters under dynamic loading have been defined. The values for the strain rate and temperature sensitivity are respectively equal to $m = \partial \log \sigma / \partial \log \dot{\epsilon} \approx 0.46$ and $v = \partial \sigma / \partial T \approx 0.63 \text{ MPa K}^{-1}$. In dynamic loading, for a strain rate close to $\dot{\epsilon} \approx 3000 \text{ s}^{-1}$, the temperature sensitivity is equal to $v = \partial \sigma / \partial T \approx 0.71 \text{ MPa K}^{-1}$.

3. Experimental impact test

3.1. Skull fracture energy

Skull fracture is a frequently observed type of severe head injury. Historically, different impact test set ups and techniques have been used for investigating skull fractures. The most frequently used are drop tower and impactor pneumatic launcher [19]. Head protection measures and proposed skull fracture criteria typically include the absorbed energy until skull fracture [19–23]. The reference of skull fracture energy ranges from 14.0 J to 68.5 J [22]. More recent studies reported values of $21.1 \text{ J} \leq E_{skull}^{fracture} \leq 40.5 \text{ J}$ at high impact velocities [23]. For comparison, the impact velocity presented by a human head assuming a

Cranial PEEK implant



Fig. 1. Cranial implant using PEEK material [6].

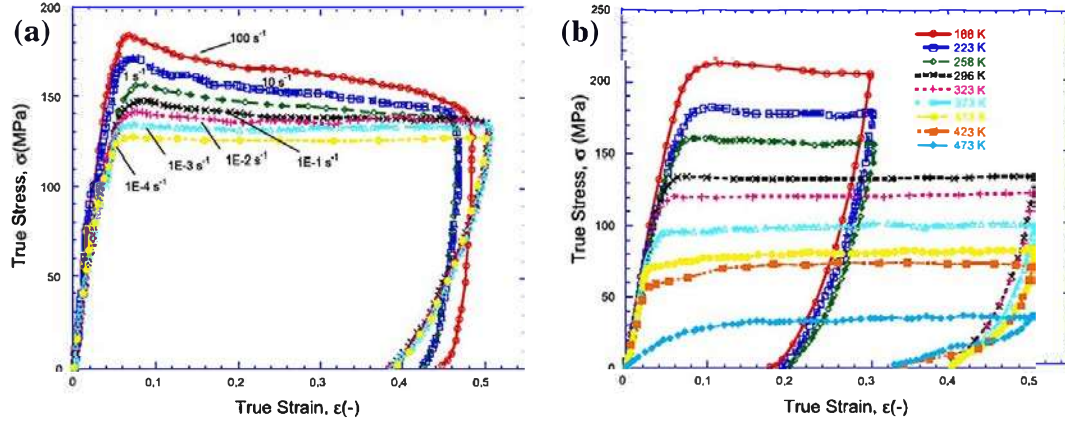


Fig. 2. Mechanical behavior of PEEK under compression for, (a) different initial strain rates at room temperature and (b) different temperatures at 0.001 s^{-1} [11].

person with a height of $h = 1.8 \text{ m}$, is equal to $V_0 = \sqrt{2gh} \approx 6 \text{ m/s}$. If we assumed an average mass of $m_{\text{head}} \approx 4.5 \text{ kg}$ for a human head, the maximum impact energy corresponding to an accidental fall is equal to $W_{\text{head}} \approx 80 \text{ J}$. In this study, the kinetic energy range has been $21.0 \text{ J} \leq E_k \leq 131.0 \text{ J}$ within the range of skull fracture energy and upper. For this proposal, perforation tests using rigid spheres have been conducted on plates of PEEK 450G and compared with Ti6Al4V titanium alloy.

3.2. Setup

The set up used was a gas gun capable of shooting a rigid spherical projectile with a mass of $m_p = 1.3 \text{ g}$ and a diameter of $\phi_p = 7.25 \text{ mm}$. This experimental device uses helium up to pressures of 200 bar to impel the projectile. The initial impact velocity V_0 was in a range of $180 \text{ m/s} \leq V_0 \leq 450 \text{ m/s}$. In order to measure the impact and the residual velocity, a high speed video camera, Photron Ultima APX RS, was used. Since the exposure time was very short, $10 \mu\text{s}$, a 1200 W HMI lamp was used to ensure adequate lighting. The camera was configured to obtain 36,000 fps. Two materials, PEEK 450G and Ti6Al4V titanium alloy were studied. The thickness of each plate was selected to obtain comparable areal density [24], a parameter frequently used to optimize impact protection, Table 2. The thickness of the Ti6Al4V was set at $t = 1 \text{ mm}$ to provide a representative comparison. Due to the boundary conditions used to avoid sliding and to ensure correct clamping of the specimen, the size of the active part of the plate was reduced to $100 \times 100 \text{ mm}^2$, Fig. 3.

4. Modeling behavior

4.1. Modeling behavior of PEEK

4.1.1. Viscoelasticity

The stress strain curves obtained at different strain rate show the viscoelastic effect observed in unfilled PEEK [11], Fig. 2. This behavior is also reported by El Qoubaa and Othman [10]. For low strain rates, the difference between elastic modulus is weak and the value obtained is constant at 3600 MPa. For high strain rates,

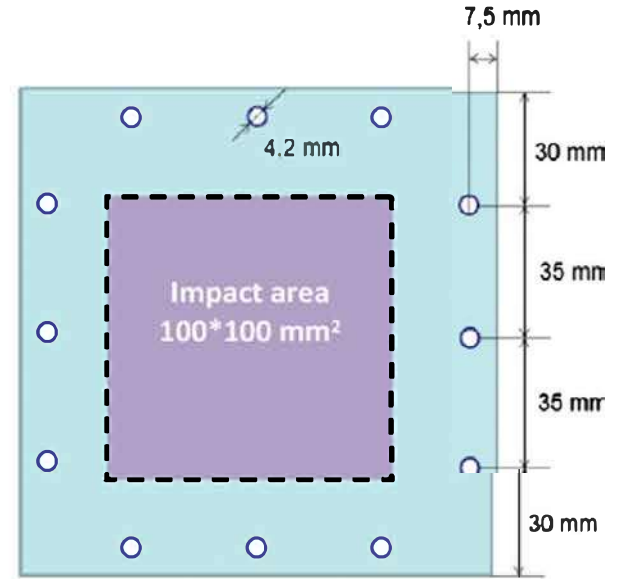


Fig. 3. Geometry of plate specimen and boundary conditions.

an increase is observed until a value of 3900 MPa for $\dot{\epsilon} \approx 10^2 \text{ s}^{-1}$ is observed according the Eq. (3), [26]:

$$E = E_0 + \eta \dot{\epsilon}^k \quad (2)$$

where E_0 is the quasi static elastic modulus, η is the consistency parameter, and k is the viscoelastic coefficient, Table 3. A direct identification of the material parameters which define the visco elastic behavior law with $E_0 = 3600 \text{ MPa}$ the static Young modulus for null strain rate.

4.1.2. Viscoplasticity

Different models have been developed with the aim of reproducing the stress strain constitutive relationship of semi crystal line polymers based on two approaches [25,26]. Firstly, a phenomenological one based on the models previously developed for metal, to introduce the viscoplasticity behavior [10,25–28]. Secondly, a physical one where the strain hardening of a semi crystal line polymer is due to entropic forces needed to orient the macromolecular chains [29,30] or by considering the intermolecular and the molecular network resistances to deformation and stretching of the chains [31]. In this work, the phenomenological Johnson Cook (JC) model [32] was used as the material model,

Table 2
Biomaterials considered for impact testing.

Material	Dimensions (mm ³)	Mass (gr)	Areal density (kg/m ²)
PEEK 450G	130 × 130 × 3	65.5	3.9
Ti6Al4V	130 × 130 × 1	78.5	4.6

Table 3

Material parameters of the viscoelastic behavior law of PEEK 450G.

E_0 (MPa)	η (MPa)	k
3600	1.25	0.9

according to strain rate and temperature sensitivity viscoplastic behavior of PEEK reported in [11], Fig. 2. Significant inelastic deformation, denoted as viscoplastic, may be observed even at very small deformation levels, Fig. 2. For semicrystalline polymers, it is reported that the viscoplastic deformation of crystalline phase is analogous to the viscoplastic deformation of crystallographic materials [33]. The JC model is frequently used in ductile metal alloys [7] and it was previously used to analyze the dynamic behavior of polymer materials [25,34]. The JC model is generally pre implemented in finite element (FE) codes, including ABAQUS/explicit [35]. This hardening law is defined by Eq. (2). The first term defines strain hardening $\bar{\epsilon}^p$, the second strain rate sensitivity $\dot{\bar{\epsilon}}^p$ and the third one is related to thermal softening Θ , Eqs. (3) and (4)

$$\sigma(\bar{\epsilon}^p, \dot{\bar{\epsilon}}^p, T) = [A + B \cdot (\bar{\epsilon}^p)^n] \left[1 + C \cdot \ln \left(\frac{\dot{\bar{\epsilon}}^p}{\dot{\bar{\epsilon}}_0^p} \right) \right] [1 - \Theta^m] \quad (3)$$

$$\Theta = \frac{T - T_0}{T - T_m} \quad (4)$$

where A and B are material constants, n is the strain hardening exponent, C the strain rate sensitivity parameter, m is the temperature sensitivity, T_0 is the initial temperature and T_m is the melting temperature. In addition, this constitutive relation coupled to the heat equation, Eq. (5), allows us to obtain the temperature increase ΔT , in adiabatic conditions:

$$\Delta T(\bar{\epsilon}^p, \dot{\bar{\epsilon}}^p, T_0) = \frac{\beta}{\rho \cdot C_p} \int_0^{\bar{\epsilon}^p} \sigma(\bar{\epsilon}^p, \dot{\bar{\epsilon}}^p, T) d\bar{\epsilon}^p \quad (5)$$

where T is the current temperature, T_0 is the room temperature, and β is the Quinney Taylor heat fraction coefficient, ρ is the density of material and C_p is the specific heat at constant pressure. The model parameters were identified according to the experimental results of compressive tests at various strain rates and temperatures [11], Fig. 2, with the methodology reported for ductile metals [7,36] and polymer [34]. First, the hardening parameters A , B and n of the JC model were identified on the basis of tests at room temperature. Parameter C was obtained by Eq. (6) at different strain rates and plastic strain equal to zero ($\bar{\sigma} = \bar{\sigma}_y$) at room temperature, 296 K. The temperature sensitivity parameter m was identified by Eq. (7) during compressive tests over with a range of imposed temperatures, Fig. 2, from initial temperature to transition temperature.

$$C = \frac{\sigma_y - A}{A \cdot \ln \left(\frac{\dot{\bar{\epsilon}}^p}{\dot{\bar{\epsilon}}_0^p} \right)} \quad (6)$$

$$m = \frac{1}{\ln \left(\frac{T - T_0}{T - T_m} \right)} \cdot \frac{\sigma}{(A + B \cdot (\bar{\epsilon}^p)^n)} \quad (7)$$

Different authors have measured the Quinney Taylor coefficient in polymers [37–39]. In this work, it is assumed constant and equal to ratio $\beta = 0.9$. This coefficient value has been reported for the inelastic work fraction generating the heat of semicrystalline polymers [40]. The identified values of JC model parameters are indicated in Table 3. Fig. 4 shows the comparison between experimental data [11] and JC model data for different strain rates at room temperature. Adequate predictions are obtained with the identified constants, Table 4. The strain rate and the temperature sensitivity of

the 450G PEEK material are reported in Fig. 5, [10,11]. As ductile metals, PEEK 450G shows non linear strain rate sensitivity and the stress increase is more significant at high strain rates than low strain rates [10]. Although this non linear strain rate of the material cannot be completely defined using the JC model due to the analytical formulation (Eq. (3)), good agreement is obtained for the identified constants from quasi static to dynamic loading. In the range of strain rates from 10^{-4} s^{-1} to 10 s^{-1} , the differences between experimental data and JC model are less than 10%. For high strain rates, from 10 s^{-1} to 10^4 s^{-1} , the differences between experimental data and JC model are less 7%.

4.1.3. Fracture model

To define completely the problem of plate impact perforation of PEEK it is necessary to introduce a fracture model. Some results have been reported in the literature concerning failure strain of unfilled PEEK [11,15] and other semicrystalline polymers such as UHMWPE [39]. In these studies, ductility of semicrystalline polymers, associated with void coalescence due tensile states, have been reported as dependent on the initial stress triaxiality σ^* ($\sigma^* = \sigma_m / \bar{\sigma}$, where σ_m is the mean stress and $\bar{\sigma}$ is the equivalent stress) and strain rate $\dot{\bar{\epsilon}}^p$. For PEEK 450G, Sobieraj et al. [9] reported an average value of failure strain of 1.1 for unnotched specimen and stress triaxiality of 0.33. However, for high triaxiality values, these authors [9] found that there was a dramatic change in the fracture micromechanism of PEEK, and the deformation and fracture micromechanics changed drastically, from one of plastic deformation and void coalescence to one dominated by crazing and brittle fast fracture with a average value of failure strain of 0.05 for notched specimens. Mourad et al. [41] reported that the ductility of semicrystalline polymer UHMWPE is strongly dependent on strain rate in such a way that fracture strain reduces with strain rate. For PEEK 450G, Rae et al. [11] found that strain failure was dependent on strain rate and temperature, with a clear upward trend in strain to failure when the temperature was increased. The fracture model used in this work has been proposed by Johnson and Cook [42] to include strain hardening $\bar{\epsilon}^p$, strain rate $\dot{\bar{\epsilon}}^p$ and temperature T dependencies. This model includes stress triaxiality σ^* , an important parameter for PEEK material. The JC frac

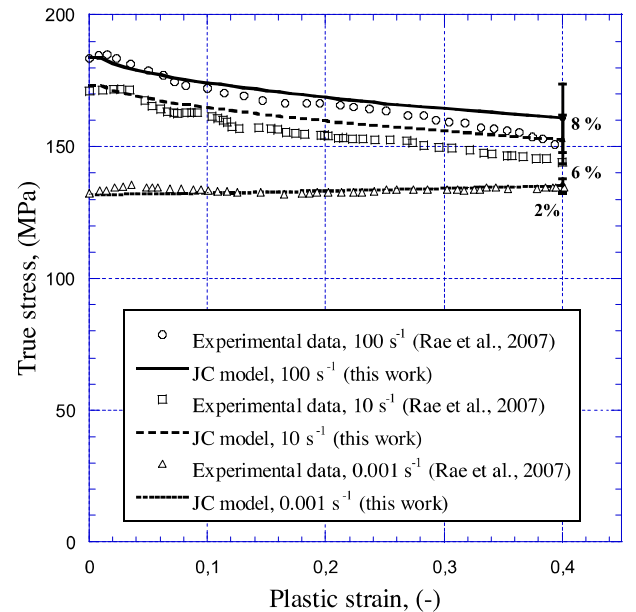


Fig. 4. Comparison of stress-strain of PEEK 450G given by the experimental data [11] and JC model predictions at room temperature and different strain rates.

Table 4

Constants used to define the thermoviscoplastic behavior of PEEK 450G at high strain rates in adiabatic conditions.

Elasticity		Thermoviscoplastic behavior					
E_0 (GPa)	ν (-)	A (MPa)	B (MPa)	n (-)	$\dot{\epsilon}_0^p$ (s ⁻¹)	C (-)	m (-)
3.6	0.4	132	10	1.2	0.001	0.034	0.7
Other physical constants		β (-)		C_0 (J/kg K)		T_m (K)	
ρ (kg/m ³)		0.9		2180		614	
1304							

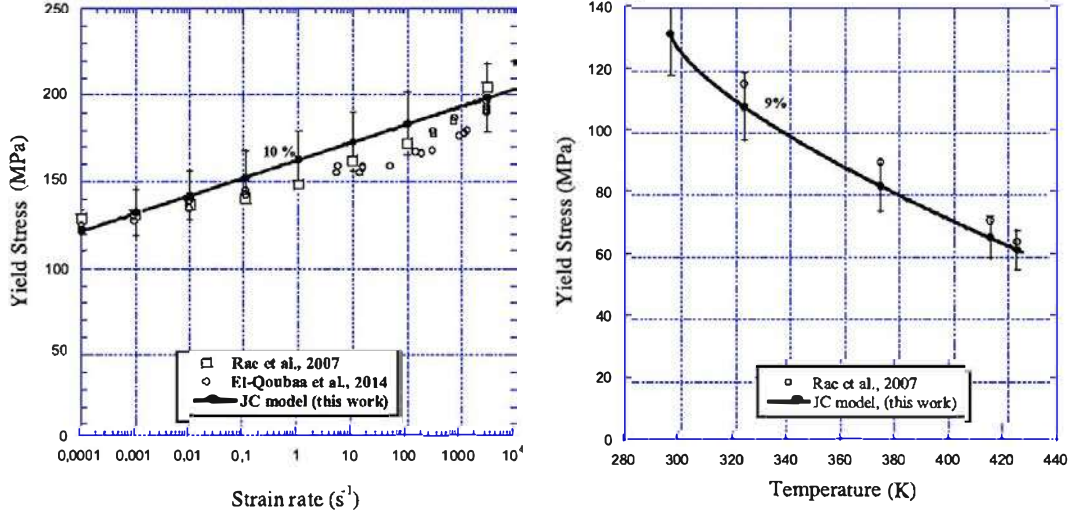


Fig. 5. Strain rate (a) and the temperature sensitivity (b) of PEEK 450G, experimental data [10,11] and JC model predictions.

ture model is frequently used in studies of ductile metal alloys [7] and it was previously used to analyze the fracture behavior of semicrystalline polymer [41]. Failure is assumed when a parameter D exceeds unity. The D parameter is summed over all increments of deformation. The evolution of D is the following:

$$D(\epsilon^p, \dot{\epsilon}^p, T) = \sum \frac{\Delta \epsilon^p}{\bar{\epsilon}_f^p(\dot{\epsilon}^p, T, \sigma^*)} \quad (8)$$

where $\Delta \epsilon^p$ is an increment of accumulated equivalent plastic strain that occurs during an integration cycle, and $\bar{\epsilon}_f^p$ is the critical failure strain level. This kind of fracture model is erosive in ABAQUS inducing an instantaneous element deletion [7] when an imposed plastic strain level is reached. Thus, using this kind of criterion the mesh in the damaged part must be very fine to not affect the numerical results in terms of energy. The plastic failure strain $\bar{\epsilon}_f^p$ is assumed to be dependent on a non dimensional plastic strain rate $\dot{\epsilon}^p/\dot{\epsilon}_0^p$, a dimensionless pressure deviatoric stress ratio σ^* and a non dimensional temperature Θ as defined previously. The dependencies are assumed separable and take the following expression:

$$\bar{\epsilon}_f^p = [D_1 + D_2 \cdot \exp(D_3 \cdot \sigma^*)] \left[1 + D_4 \cdot \ln \left(\frac{\dot{\epsilon}^p}{\dot{\epsilon}_0^p} \right) \right] [1 + D_5 \cdot \Theta] \quad (9)$$

where D_i are failure constants. The constant D_1 implies a finite strain to fracture even at very high values of stress triaxiality [41]. This value is assumed to be $D_1 = 0.05$, according an average failure strain of 0.05 for high triaxiality reported for PEEK 450G [9]. The constant D_2 and D_3 are identified from experimental data [9] often site tests of unnotched specimens of PEEK 450G at strain rate equal to 0.001 s⁻¹ and room temperature. The value is close to $\bar{\epsilon}_f^p \rightarrow 1.1$ in these conditions. The constant D_4 and D_5 are identified from experimental data [11] of uniaxial tests of unnotched specimens of PEEK

450G at different strain rate and temperatures. According these experimental data D_4 defines the dependence on the strain rate as negative sensibility whereas D_5 defines the temperature as large positive sensibility, Fig. 6. The constants identified of fracture model of PEEK 450G are given in Table 5.

4.2. Modeling behavior of Titanium Ti6Al4V

Ti6Al4V titanium alloy is a biocompatible material with high yield stress, $\sigma_y = 1098$ MPa and has been frequently used for medical applications [1]. This material has been widely studied with reports of the plasticity and damage models available in the literature [43–45], including strain rate and temperature dependence. To define its thermoviscoplastic behavior several approaches may be used including physical [43] and phenomenological formulations [44,45]. In this work, Johnson Cook hardening law [32], Eq. (3) and JC fracture model [42], Eq. (9), have been used to describe the titanium alloy behavior drawing on reported data [45], Tables 6 and 7.

5. Numerical model

A Lagrangian 3D finite element model for the simulation of the perforation process was developed in ABAQUS/explicit [35]. The geometry of plates is equal to the active area of the experimental tests specimens (100 × 100 mm) with a value of thickness of 1 mm for Ti6Al4V and 3 mm for PEEK (Table 2). The fully 3D configuration allows the model to describe the radial cracking and the petalling failure mode that characterize the perforation of plates by spherical projectiles [46]. The target mesh developed is shown in Fig. 7, where twelve elements were placed across the thickness of the target. This is in agreement with the recommendations

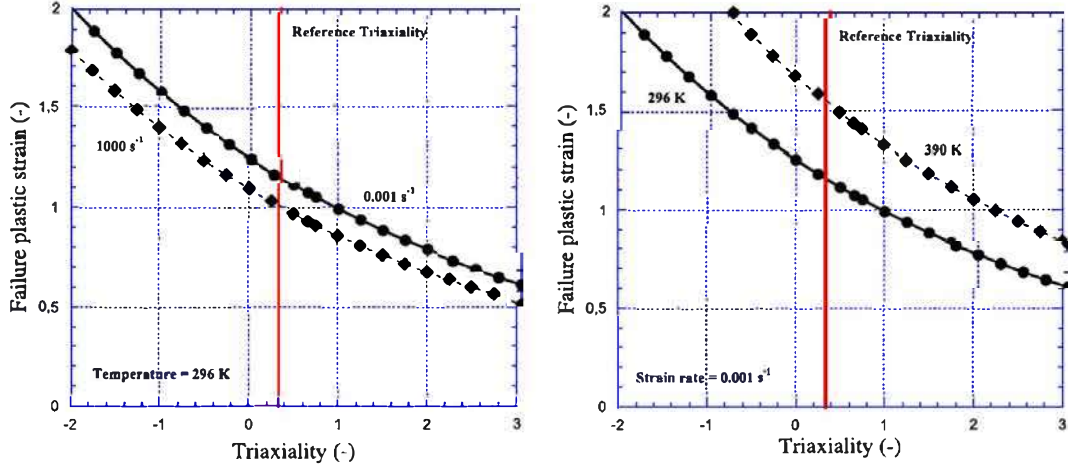


Fig. 6. Failure strain level (JC fracture model) depending on solicitation state at different strain rates (a) and at different temperatures (b).

Table 5
Constants used to define fracture model of PEEK 450G.

Fracture criterion constants				
D_1	D_2	D_3	D_4	D_5
0.05	1.2	-0.254	-0.009	1.0

Table 7
Constants used to define fracture model [44].

Fracture criterion constants				
D_1	D_2	D_3	D_4	D_5
-0.09	0.27	0.48	0.014	3.87

reported [35], where it is suggested that at least four elements should be used through the thickness when modeling any structures carrying bending loads. The mesh presents radial symmetry to avoid appearance of spurious generation of cracks. The mesh is split into three different zones. The zone directly affected by the impact has been meshed with 32,400 tri linear elements with reduced integration (C3D8R in ABAQUS notation). In order to reduce the computational time, beyond the zone directly affected by the impact is defined a transition zone using 48,000 elements. After the transition zone, the mesh is defined using C3D8R elements until reaching the perimeter of the target. This optimum configuration has been obtained from a convergence study using different mesh densities. Since the experimental observations revealed absence of erosion on the projectile surface after the impact (the projectile was not deformed plastically in any test), the projectile has been defined as rigid body. This enables a reduction in the computational time required for the simulations. A constant friction coefficient value $\mu = 0.2$ has been used to define the contact projectile/plate [47]. The potential dependence of the friction coefficient on the temperature and the sliding velocity is not taken into account. The constant value used for the friction coefficient is based on the assumption of a constant pressure along the projectile plate contact zone. The authors confirmed this hypothesis by FE analysis of different projectile target configurations [45].

Table 6
Constants used to define the thermoviscoplastic behavior of Ti6Al4V at high strain rates in adiabatic conditions [45].

Elasticity		Thermoviscoplastic behavior					
E (GPa)	ν (-)	A (MPa)	B (MPa)	n (-)	ϵ_0^p (s ⁻¹)	C (-)	m (-)
109.8	0.31	1098	1092	0.93	1.0	0.014	1.1
Other physical constants							
ρ (kg/m ³)			β (-)	C_p (J/kg K)		T_m (K)	
4428			0.9	560		1878	

The impact velocities covered with the numerical simulations are those covered during the normal impact experiments.

6. Results and discussion

6.1. Energy absorption and residual velocity

Firstly, the experimental results of impact velocities are analyzed. Fig. 8a shows the residual velocity versus impact velocity (V_r , V_0) curves obtained for both materials tested, PEEK 450G and Ti6Al4V. The ballistic limit V_{bl} is the maximum value of the initial impact velocity V_0 which induces a residual velocity V_r equal to zero. The ballistic limit of PEEK 450G, $V_{bl}^{PEEK} \approx 265$ m/s was found greater than that corresponding to the Ti6Al4V plates, $V_{bl}^{Ti6Al4V} \approx 232$ m/s. The results shown in Fig. 8 have fitted via the expression proposed by Recht and Ipson [48]:

$$V_r = (V_0^k - V_{bl}^k)^{1/k} \quad (10)$$

where k is a fitting parameter. The values of k determined are $k = 1.9$ for PEEK 450G and $k = 2$ for Ti6Al4V. The residual velocity of PEEK 450G plates within the range of impact velocities tested is lower than Ti6Al4V. It was noted that the flow stress of Ti6Al4V is ten times higher than the flow stress of PEEK 450G in quasistatic conditions [11,45], and seven times in dynamic conditions,

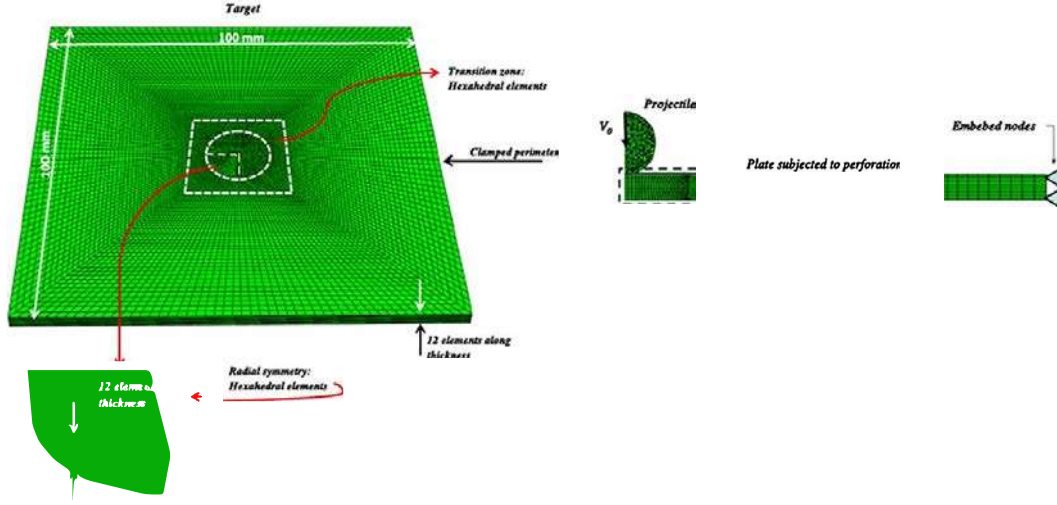


Fig. 7. Numerical configuration used in the simulations.

10^3 s^{-1} to 10^4 s^{-1} , [10,11]. This suggests that the energy absorption mechanics shown by both materials investigated are largely different to each other. Thanks to the measurements described previously, it is possible to estimate the energy absorption W of PEEK 450G material under dynamic impact by Eq. (11) and the minimum energy to perforation, $W_{\text{perforation}}$, Eq. (12)

$$W = \frac{1}{2} m_p (V_0^2 - V_r^2) \quad (11)$$

$$W_{\text{perforation}} = E_{\text{fracture}}^{\text{skull}} \quad (12)$$

Fig. 8b shows the kinetic energy of the projectile converted into energy absorption W of plate. For both materials W increases with initial velocity. This tendency is in agreement with experimental results published in the literature for spherical noses of projectile [7]. Comparing the values obtained, it is observed that PEEK material can absorb enough energy to avoid skull injury independent of the impact velocity, Fig. 8b. The reference used for comparison is the maximum skull fracture energy, $E_{\text{fracture}}^{\text{skull}} = 40.5 \text{ J}$, reported in [23]. The values of perforation energy are respectively $W_{\text{perforation}}^{\text{PEEK}} = 45.6 \text{ J}$ and $W_{\text{perforation}}^{\text{Ti6Al4V}} = 35 \text{ J}$. Moreover, in the full range of impact velocities, PEEK material is more efficient for energy absorption compared to titanium alloy with a medium ratio of

$R = 1.26$. One of the reasons for the good capability to absorb energy is the adiabatic deformation of PEEK at high strain rates with large strain values ($\epsilon > 1.0$). This behavior is characteristic of compressive states in dynamic process, as reported Rae et al. [11] for Taylor impact tests.

6.2. Failure mode

6.2.1. Ductile behavior of PEEK

Figs. 9 and 10 illustrate the final stage of the impact process for different initial velocities and both materials tested. The failure mode of PEEK plates is clearly different from that observed in titanium alloy. The shear failure of Ti6Al4V plates is characterized by small energy consumption. Namely, the energy required for perforation is that required for the onset of the shear band, plug ejection and cracks propagation, Fig. 10. The length of cracks is larger at velocities below the ballistic limit, and the crack propagation path is transverse to the rolling direction [49], Fig. 10. Once the instability is formed, a very low energy consumption is needed to perforate the plate. However, at all impact energies, ductile process of PEEK plates was noted and no evidence of brittle failure was observed, Figs. 9 and 11. Moreover, multi hit capability [50] for impact absorption energy of PEEK material has been demonstrated due to localized ductile damage, Fig. 11b. Based on the ballistic

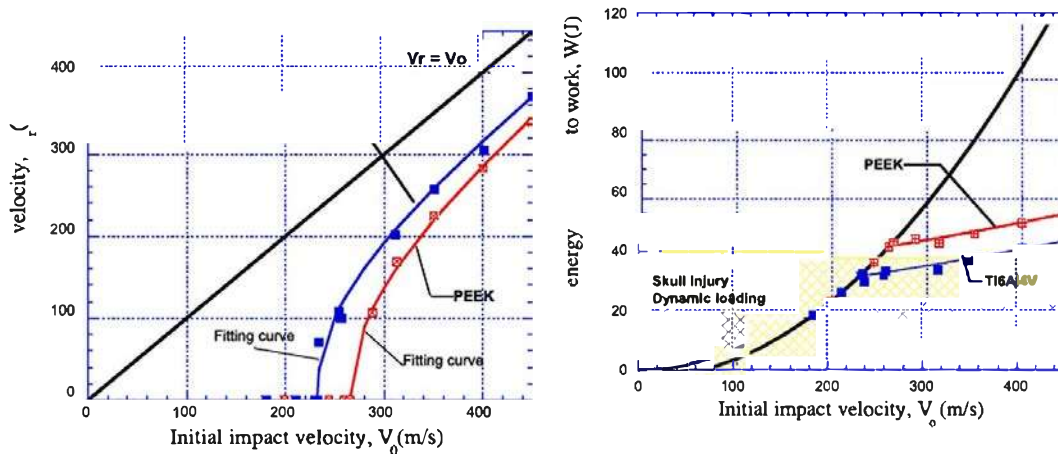


Fig. 8. (a) Residual velocity V_r versus impact velocity V_0 , comparison between PEEK 450G and Ti6Al4V; (b) energy absorbed the plate W versus impact velocity V_0 , comparison between PEEK 450G and Ti6Al4V.

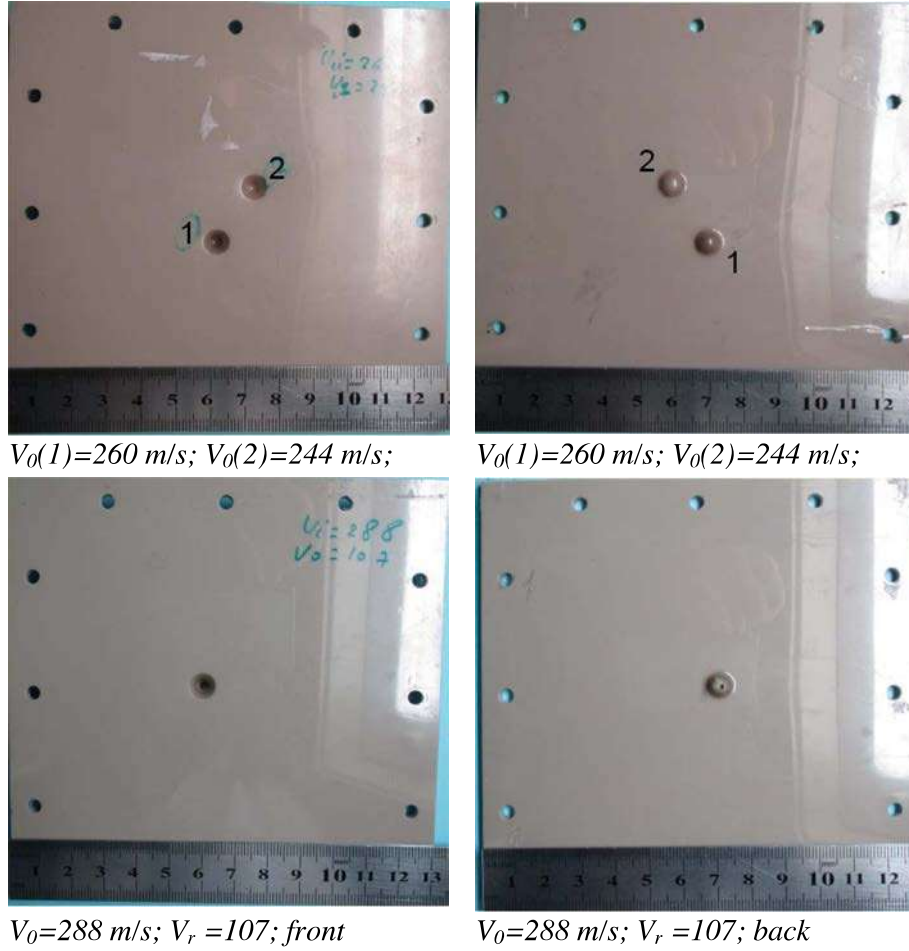


Fig. 9. Final stage of the perforation process of PEEK for different impact velocities (front and back of plates).

tests, it is observed a ductile failure mode without plug ejection as frequently observed for metal materials using a spherical or hemi spherical projectile [7,46].

Comparing PEEK with metals, a hole enlargement is observed when the projectile is passing through the plate followed by an enclosing hole stage, Fig. 12. A local deflection is also observed without real bending of the whole plate compared to metal. The plastic deformation is confined close to the impact zone, with a maximum value of two times the diameter of the projectile. Moreover, using SEM photo, it is possible to note a ductile failure mode and large plastic deformation lines, Fig. 13. The same failure mode has been observed independent of the initial impact velocity V_0 used. This observation agrees with the results provided by Rae et al. using Taylor's test at high impact velocity [11].

6.2.2. Darkening at large strain

At all velocities, darkening is noticed in the highly deformed regions associated to local damage in plates. Fig. 14 clearly shows the concave rod end and the discolouration associated with the large strain regions. A number of authors have focused on identifying whether the colour change of PEEK was a result of the strain rate or large strain. Fig. 14 shows the cross sections of virgin and three large strain samples impacted at different velocities below ballistic limit and photographed under identical conditions. The more impact energy, the larger the strain and darker the damage area. From the associated colour changes it is therefore clear that the colour change is associated with large strain compression,

not strain rate. PEEK in tension undergoes stress whitening in common with many other polymers.

6.2.3. Crystallinity at high strain rate and large strain

The percentage of crystallinity is responsible for the material fragility. To change the microstructure from amorphous to crystal line, the material needs additional energy mainly derived from temperature increase ΔT . For this reason, it is interesting to analyze the behavior of PEEK polymer at high velocity impact V_0 to check if a brittle transition is observed since for high velocity, the initial temperature T_0 shows a strong increase. Differential scanning calorimetry (DSC) experiments were carried out in a Perkin Elmer Diamond calorimeter with nitrogen as the purge gas. PEEK samples of highly deformed regions corresponding to different impact energies of tests have been analyzed. Samples with a mass of $\approx 10 \text{ mg}$ were sealed in a 50 ml aluminium pan, and an empty pan was used as reference. They were heated from 293 K to 433 K at a heating rate of 10 K/min and then cooled at the same rate. Enthalpies of melting were determined with an average value of $\Delta H_f \approx 39 \text{ J g}^{-1}$. The crystallinity of all samples deformed to large strains was found similar and equal to 30%. There is a point of contention in this regard about the relationship between high strain rate, large strain and crystallinity. While some authors, Hamdan and Swallowe [12] reported an increase in crystallinity in dynamic conditions under high strain rates, $\dot{\epsilon} \approx 10^3 \text{ s}^{-1}$, others authors reported decrease of crystallinity in dynamic conditions for samples deformed to large strains $\epsilon > 0.1$ [11]. An explanation of this

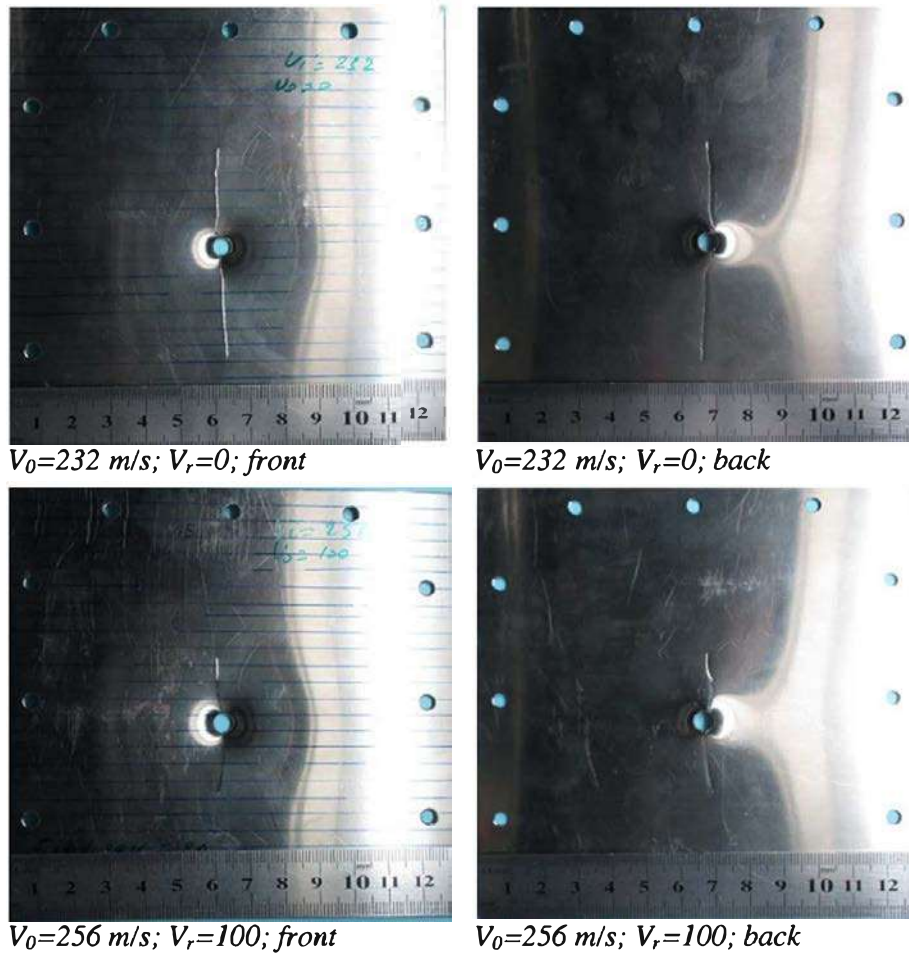


Fig. 10. Final stage of the perforation process of Ti6Al4V for different impact velocities (front and back of plates).

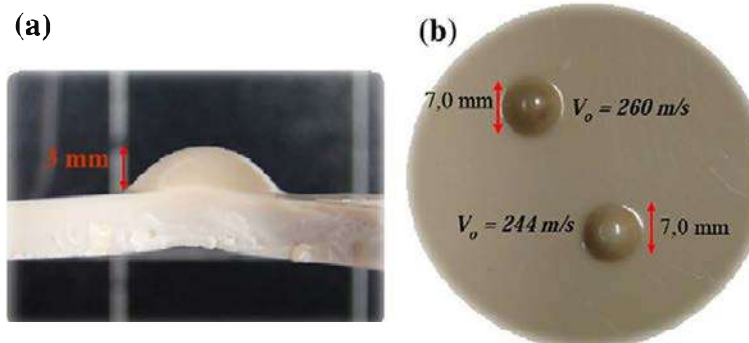


Fig. 11. Final stage of the impact process of PEEK at impact velocity below ballistic limit: (a) $V_0 = 200 \text{ m/s}$; (b) two different impact at $V_0 = 260 \text{ m/s}$ and $V_0 = 244 \text{ m/s}$.



Fig. 12. Final stage of PEEK at impact velocity above ballistic limit: $V_0 = 288 \text{ m/s}$ ($V_r = 107 \text{ m/s}$).

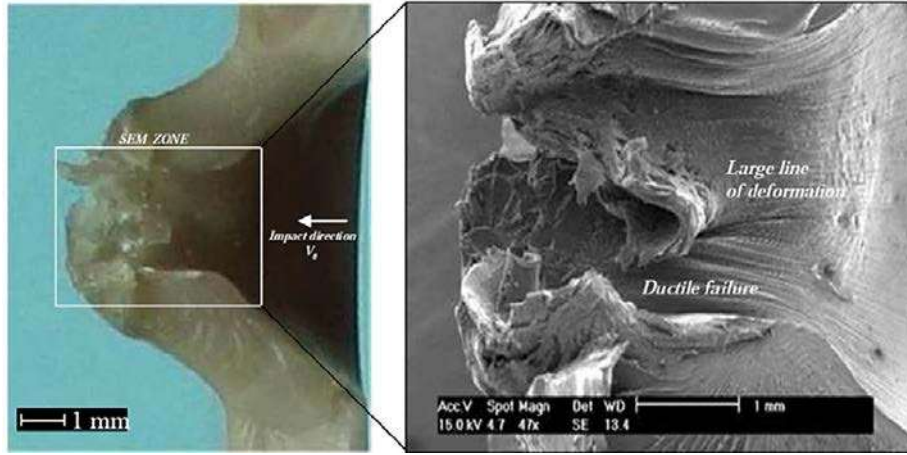


Fig. 13. Cross section of final stage of PEEK at impact velocity $V_0 = 288$ m/s ($V_r = 107$ m/s).

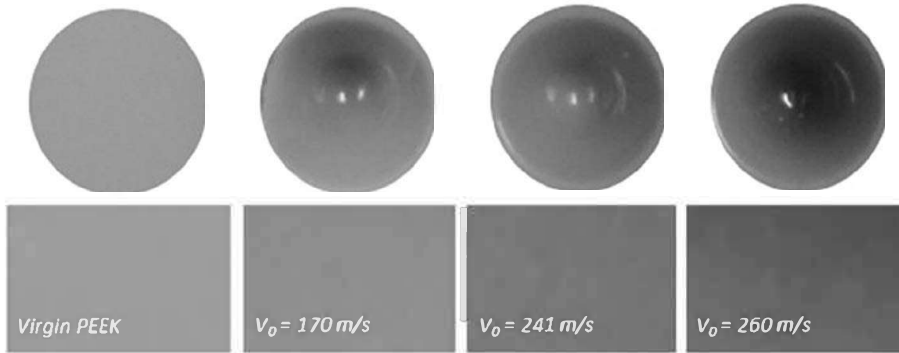


Fig. 14. Polished sections of four PEEK samples photographed under identical conditions to show the color change associated with large-strain deformation. The grayscale value, in the range from 0 to 255 where 255 is black, 93 is for virgin PEEK, 114 for $V_0 = 170$ m/s, 141 for $V_0 = 241$ m/s and 156 for $V_0 = 260$ m/s.

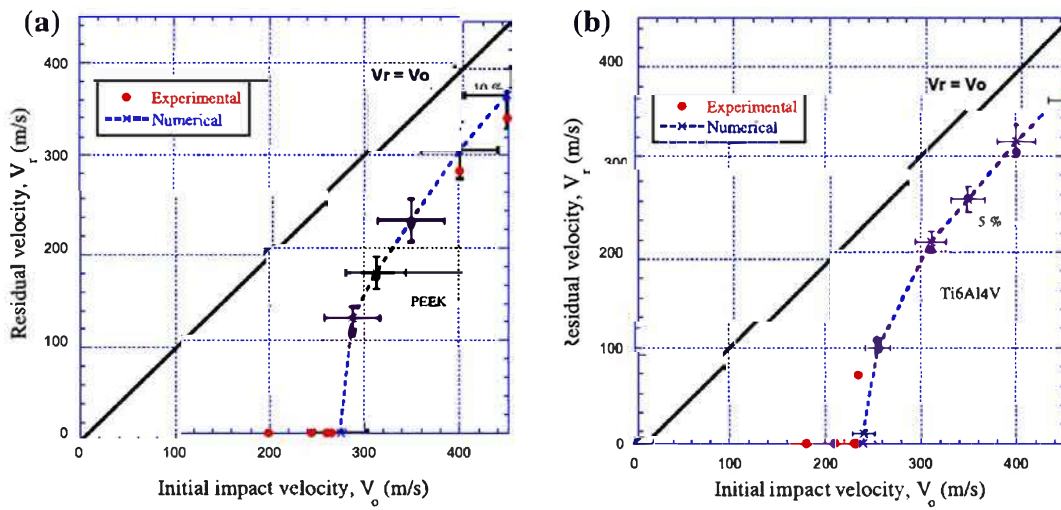


Fig. 15. Experimental and numerical data of residual velocity versus impact velocity: (a) PEEK 450G, (b) Ti6Al4V.

behavior is not obvious. In the research reported here, no changes in degree of crystallinity in relation with virgin PEEK were measured. Below glass transition temperature, crystallinity evolution can be considered as a competition between the stabilizing effect

of strain rate on crystal structure [51] and the probable destruction associated with large strain. In this regards, it has been reported that high compressive deformation can degrade the crystalline structure of polymer at temperatures $T < T_g$ [52].

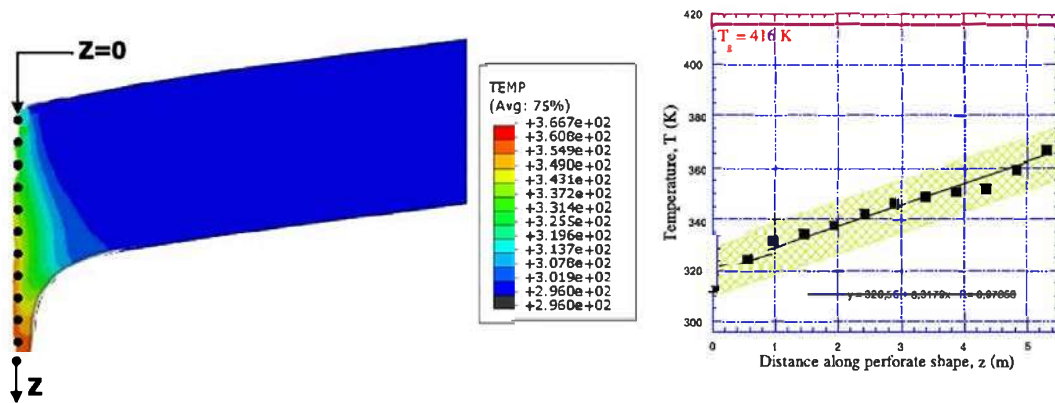


Fig. 16. Numerical predictions of maximum temperature increase in PEEK for impact velocity, $V_0 = 265$ m/s.

6.3. Numerical simulation

A good correlation was found between experimental and numerical results, less than 10%, which demonstrates that the models used in this study faithfully reproduce the impact behavior of tested materials, PEEK 450G and Ti6Al4V, Fig. 15. From numerical simulations, the maximum temperature increase of PEEK on the contact zone specimen ball has been computed. Therefore, an estimation of the temperature during the process of plastic deformation has been performed for a velocity corresponding near to the ballistic limit $V_0 = 265$ m/s, Fig. 16. An increase in the temperature is observed during perforation process. However, for high impact velocity the temperature appears to be lower. The reason is certainly due to the bending effect which induces an increase in plastic work. For high impact velocity the failure appears quickly without bending effect and in this case the plastic work is lower. Generally, the tendency of the temperature is to increase with an average value corresponding to complete failure equal to $\Delta T \approx 100$ K. The maximum local temperature has been compared with the glass transition temperature and the melting temperature. It is observed that locally the temperature is lower than the glass transition temperature, Fig. 16. This numerical prediction is in agreement with experimental data, where it is observed that the material is still ductile for a temperature $T_0 < T_g = 416$ K [11]. Indeed at high impact velocity, no brittle behavior has been observed and PEEK material is capable to absorb more energy as velocity increases, Fig. 8b. Thus it is observed using these numerical simulations that the behavior of the material must be well defined since it depends strongly on the temperature sensitivity and strain rate and amplified by large deformation. During this kind of application the strain level observed is higher to 1.

7. Conclusions

The impact behavior of unfilled PEEK was examined through a combination of experiments and finite element simulations. In the full range of impact kinetic energies considered, from 21 J to 131 J, PEEK material is more efficient for energy absorption in comparison with titanium alloy. During experiments, the material was observed to behave in a ductile manner without evidence of brittle failure and darkening is noticed in highly deformed regions. Finite element simulations of experiments were performed using a plasticity model typically used for ductile metals. The simulation was in good agreement with the experimental data, confirming the predominantly ductile response of PEEK under high strain rate. From numerical simulations, it has been computed the maximum temperature increase in the perforation process. The numerical predictions of maximum local temperature are lower than the glass

transition temperature of PEEK, in agreement with the ductile behavior observed in impact tests. In conclusion, PEEK appears to be an attractive candidate as matrix material for impact applications, implants or to design passive security equipment such as helmets.

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